
input
NHWC, fp16
im2col
$k_h k_w$ strided copies
into a padded buffer
costs about as much
as the sparse GEMM
spatha.spmm
packed V:N:M weights
Sparse Tensor Cores
K padded to $\max(128, 8M)$,
$\leq$65,535 column tiles per call
output
$(BH_oW_o) \times C_{\text{out}}$
= NHWC
view, no copy
BN, activation,
pool, sum/concat
(dense, un-
changed)
58–66% of forward
time; not accelerated
Timed variants (interleaved rounds, median of 5): dense cuDNN · dense-best (per layer
faster of cuDNN / im2col+GEMM, the baseline) · all-Spatha · hybrid (per layer fastest)